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Cardiac resynchronisation as a rescue therapy in patients with catecholamine-dependent overt heart failure: Results from a short and mid-term study

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Abstract

Background: Cardiac resynchronisation therapy (CRT) is a validated treatment for heart failure (HF) patients in NYHA class III–IV despite optimal medical therapy. We aimed to assess the beneficial effects of CRT in patients with catecholamine-dependent overt HF (CDOHF).
Methods: We studied 20 CDOHF patients who had undergone CRT implantation. Patients had a mean baseline QRS duration of 174±25 ms and/or echocardiographic asynchrony, and LVEF of 18±3%. Mean follow-up was 18±12 months. Dependence on catecholamine agents was defined as the inability to stop or reduce drug infusion without re-occurrence of hypotension, low urine output and hypoxaemia.
Results: After CRT implantation, catecholamine agents were mostly withdrawn within 2 days and blood pressure, urine output and BNP rapidly improved within 24 h. During follow-up, survival rates were 85% at 3 months, 80% at 6 months and 55% at 18 months. Among the 9 deaths, 5 were related to overt HF, 3 to sudden cardiac death and 1 to non-cardiac death. LVEF improved from 18±3% to 21±4% three months after CRT implantation.
Conclusion: "Rescue" CRT implantation in CDOHF patients allowed a rapid and successful catecholamine weaning in all studied patients. Furthermore, this immediate beneficial effect is sustained for more than one year in surviving patients.
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Keywords: CRT; Rescue; Acute decompensated heart failure

1. Introduction

Cardiac resynchronisation therapy (CRT) has become an effective treatment in patients in New York Heart Association (NYHA) class III–IV with impaired left ventricular ejection fraction (LVEF,<35%) and wide QRS and/or echocardiographic ventricular dyssynchrony, despite optimal medical therapy [1–7]. Furthermore, the Cardiac Resynchronization — Heart Failure (CARE-HF) trial has recently demonstrated a significant improvement in survival in patients with a CRT device [8].

However, as end-stage heart failure (HF) patients, and especially those with catecholamine-dependent overt HF (CDOHF), are excluded from CRT clinical trials, the impact of CRT implantation in this population remains unknown. Although ventricular assist devices (VADs) and/or immediate cardiac transplantation are recommended in CDOHF patients, their use is limited by associated severe organ dysfunction or advanced age. The effects of CRT were originally

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tested in acute haemodynamic studies [9–14]. Early observations showed that atrioventricular resynchronisation with shorter atrioventricular delay decreased both mitral and tricuspid regurgitations, increased diastolic left ventricular filling time and improved cardiac output [9,10]. Furthermore, ventriculo-ventricular resynchronisation acutely enhanced both systolic and diastolic cardiac functions with improvement of the cardiac index and pulmonary capillary wedge pressure [11–14].

Our hypothesis was that CRT implantation may induce a rapid and possibly sustained haemodynamic benefit in patients with CDOHF and wide QRS. To test this, we set out to assess both the immediate outcomes – including haemodynamic improvement, weaning from catecholamine and improvement in biomarkers – and the mid-term outcomes of CRT implantation in CDOHF patients with wide QRS and/or echocardiographic ventricular asynchrony for whom no other medical or surgical alternative was available.

2. Methods

2.1. Patient population

Between January 2002 and September 2006, 225 patients in our two institutions were routinely implanted with a CRT device for overt HF due to either ischaemic or idiopathic cardiomyopathy according to published recommendations [15]. Among these 225 patients, 20 had been hospitalised for end-stage HF and were dependent on catecholamine infusion. These 20 CDOHF patients represent our study group and were followed for more than one year.

CDOHF patient status was defined as the recurrence of both clinical and biological signs of low cardiac output despite 3 attempts of very progressive weaning from catecholamine agents (i.e. decrease of 1 μ/kg/min daily down to 2 μ/kg/min, followed by a decrease of 0.1 μ/kg/min every 8 h) with optimised preload and oral vasodilator therapy (such as angiotensin-converting enzyme (ACE) inhibitors) as recommended [16]. Clinical signs of low cardiac output were defined as persistent low blood pressure (systolic blood pressure < 90 mmHg), hypoxaemia, tachycardia, mottled skin, cold extremities, cyanosis, neurologic disorders, low urine output with renal failure (less than 0.5 ml/kg/h) and liver failure. In our centres, CDOHF patients are candidates for left VAD and/or heart transplant unless they are advanced in age or have severe associated organ dysfunction. The reasons why the 20 patients included in our study were ineligible for left VAD or heart transplantation were: 12 were over 70 years old with or without severe non-cardiac diseases; of the remaining eight, two patients had severe chronic lung disease, two had severe chronic renal failure, two had severe diabetes with major vascular and renal complications and two had rejected the option of heart transplantation. We thus approached these patients with the option of a CRT implantation after informing them of the severity of their clinical situation and after receiving ethics committee approval. All patients underwent CRT implantation within 48 h of the last attempt to wean from catecholamine. All patients gave their written informed consent.

Systolic and diastolic blood pressure and urine output were recorded just before and after CRT implantation. The following biomarkers were measured 24 h before and the day following CRT implantation: brain natriuretic peptide (BNP) (normal < 100 pg/ml); uraemia (normal < 7 mmol/l) and creatinine (normal < 120 μmol/l). Reintroduction of ACE-inhibitors and beta-blockers, which had been stopped when the patients were admitted to the intensive care unit (ICU) due to their CODHF status, was attempted as previously described for the weaning from catecholamines.

Before CRT implantation, patients with a QRS interval of less than 160 ms were required to meet two of three additional echocardiographic criteria for asynchrony described in the "CARE-HF" study [8]: an aortic preejection delay of more than 140 ms, an interventricular mechanical delay of more than 40 ms, or delayed activation of the postero-lateral left ventricular wall.

2.2. Implantation of CRT devices

All the patients received light sedation and oxygen throughout the procedure and were under permanent anaesthesiologist supervision. Blood pressure and central venous pressure was permanently monitored using intra-arterial and intravenous catheters. Intravenous catecholamine agents were continued at the same doses during the whole procedure.

All leads were implanted transvenously. The atrial lead (in patients in sinus rhythm) was placed high in the right atrium. The left ventricular lead was optimally positioned within the coronary sinus after studying right and left anterior oblique views obtained from a venogram performed during the procedure. The target site was preferably the lateral wall, midway between base and apex, but other postero- or antero-lateral veins were chosen when this vein was absent. If the lateral veins were not accessible, the great cardiac vein or the middle cardiac vein was used. The right ventricular lead was positioned in all patients as far as possible from the left ventricular lead. The CRTs used were either a standard dual-chamber technology devices with built-in adapters to synchronize the pacing of the two ventricles or current CRT devices with separate exits for each lead.

2.3. Post-implantation period

All patients remained in the ICU until the intravenous medications could be withdrawn. Permanent haemodynamic monitoring by intra-arterial catheter was discontinued as soon as the clinical and biological signs of unstable HF improved. Dobutamine, dopamine or epinephrine were then progressively discontinued as the haemodynamic status of the patient improved along with clinical improvement. Patients were then discharged from the ICU and transferred

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Table 1
Baseline patient characteristics before CRT implantation

Characteristic	CDOHF CRT (n=20)
Age (year)	67±10
Male n (%)	18 (90)
NYHA IV n (%)	20 (100)
Dilated cardiomyopathy n (%)	8 (40)
Ischaemic heart disease n (%)	12 (60)
Systolic blood pressure (mm Hg)	84±7
Diastolic blood pressure (mm Hg)	55±5
LVEF (%)	18±3
LVEDD (cm)	7.6±0.8
QRS duration (ms)	174±25
Use of ACE inhibitor or ARB n (%)	**0 (0)**
Use of beta-blocker n (%)	**0 (0)**
Use of spironolactone n (%)	10 (50)
Use of loop diuretic n (%)	20 (100)

All quantitative variables are expressed in mean±standard deviation. *$p<0.05$.
CDOHF=catecholamine-dependent overt heart failure, NYHA=New York Heart Association, LVEF=left ventricular ejection fraction, LVEDD=left ventricular end-diastolic dimension, ACE=angiotensin-converting enzyme, ARB=angiotensin-receptor blocker.
Bold characters: ACE inhibitor and beta-blocker were discontinued because of unstable heart failure with clinical and biological signs of low cardiac output at the time of CRT implantation.

to a regular cardiac unit and returned home in the absence of complications or recurrence of unstable HF. Before hospital discharge, optimisation of both the atrio-ventricular and ventriculo-ventricular intervals for the more recent CRT device with separate exits, was performed in all patients. Patients returned to the centres for clinical assessment and device checking at one month and then every three months.

2.4. Statistical analysis

Data of continuous variables are presented as mean± standard deviation. Data before and after CRT implantation were compared using paired Student t test; $p<0.05$ was considered as statistically significant.

3. Results

3.1. Study population

Our study population characteristics are summarized in Table 1. Patients were predominantly male with a mean age of 67±10 years. Twelve patients had ischaemic and eight non-ischaemic cardiomyopathy. The mean QRS duration was 174±25 ms and the mean LVEF 18±3%. Fifteen patients were in sinus rhythm while the remaining five were in permanent atrial fibrillation. All the patients had wide QRS duration (>130 ms) at the time of implantation. Transthoracic echocardiography with assessment of ventricular asynchrony was performed in 12 patients with a QRS of between 130 and 160 ms. Of these 12 patients, seven had all three echocardiographic criteria for asynchrony, three had an aortic preejection delay of more than 140 ms and an interventricular mechanical delay of more than 40 ms, and two had an interventricular mechanical delay of more than 40 ms and a delayed activation of the postero-lateral left ventricular wall. All patients had been hospitalised at least twice within the 15 months prior to CRT implantation for decompensated HF. In half of these previous episodes, dobutamine infusion had already been added concomitantly to routine medication to improve the haemodynamic status.

Table 2
Baseline characteristics of CDOHF patients before CRT implantation

Patient number	Rhythm	Age	Sex	Heart disease	LVEF	QRS width	DHF episodes prior to CRT	Inotropic agent	Dose
1	SR	78	M	Ischaemic	15	180	3	Dobutamine	10 μ/kg/mn
2	SR	74	M	Ischaemic	20	140	2	Dobutamine	7.5 μ/kg/mn
3	SR	68	M	Dilated	20	140	3	Dobutamine	5 μ/kg/mn
4	SR	60	F	Dilated	20	180	2	Dobu+epinephrine	2.5 μ/kg/mn+0.5 mg/h
5	SR	64	F	Dilated	25	160	3	Dobutamine	5 μ/kg/mn
6	AF	75	M	Ischaemic	25	150	2	Dobutamine	10 μ/kg/mn
7	AF	70	M	Dilated	20	150	2	Dobutamine	5 μ/kg/mn
8	SR	64	M	Ischaemic	15	180	2	Dobutamine followed by Levosimendan	10 μ/kg/mn 0.4 μ/kg/mn
9	SR	73	M	Ischaemic	17	180	3	Dobutamine	10 μ/kg/mn
10	SR	75	M	Ischaemic	15	200	2	Dobutamine	7.5 μ/kg/mn
11	AF	77	M	Dilated	20	200	2	Dobutamine	5 μ/kg/mn
12	AF	50	M	Dilated	20	140	3	Dobutamine	5 μ/kg/mn
13	AF	43	M	Dilated	17	140	3	Dobu+dopamine	10 μ/kg/mn+3 μ/kg/mn
14	SR	50	M	Ischaemic	10	200	2	Dobutamine	10 μ/kg/mn
15	SR	74	M	Ischaemic	15	180	3	Dobutamine	5 μ/kg/mn
16	SR	75	M	Ischaemic	15	180	3	Dobutamine	7.5 μ/kg/mn
17	SR	75	M	Ischaemic	15	200	2	Dobutamine	5 μ/kg/mn
18	SR	77	M	Dilated	15	190	3	Dobutamine	7.5 μ/kg/mn
19	SR	63	M	Ischaemic	20	170	3	Dobutamine	5 μ/kg/mn
20	SR	70	M	Ichaemic	20	170	3	Dobutamine	5 μ/kg/mn

HR=heart rhythm, SR=sinus rhythm, AF=atrial fibrillation, M=male, F=female, DHF=decompensated heart failure; N/A=not available.

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Table 3
Haemodynamic, electrical and biological measures before and after CRT implantation

Parameters	Before CRT implantation	24-hours after CRT implantation	*p*
SBP (mmHg)	84±7	99±10	<0.0001
DBP (mmHg)	55±5	60±6	<0.001
Urine output (ml/day)*	817±264	2256±1550	<0.01
Dobutamine (μ/kg/min)	7.5±25	0†	<0.0001
QRS duration (ms)	174±25	163±18	<0.01
BNP (pg/ml)‡	2176±863	960±520	<0.001
Uraemia (mmol/l)§	12±5	9±4	<0.001
Creatinine (μmol/l)§	185±42	153±34	<0.001
LVEDD (cm)	7.6±0.8	7.5±0.9	NS
LVESD (cm)	6.5±0.8	6.3±0.8	NS
LVEF (%)	18±4	21±5†	0.004

All quantitative variables are expressed in mean±standard deviation.
SBP=systolic blood pressure; DBP=diastolic blood pressure; BNP=brain natriuretic peptide; LVEDD=left ventricular end-diastolic dimension; LVESD=left ventricular end-systolic dimension; LVEF=left ventricular ejection fraction.*n=12; ‡n=13; §n=13.†Dobutamine was stopped within 48 h in 18/20 patients and within 5 days for 2. Post-implantation LVEF were measured between 1 to 3 months after CRT implantation.

At the time of CRT implantation, all patients were on dobutamine infusion (mean dose: 7.5±2.5 μ/kg/min); dobutamine was combined with epinephrine in one patient, with levosimendan in another and with dopamine in a third. Individual baseline data for the 20 patients are listed in Table 2. The haemodynamic status and biological values for the 20 CDOHF patients, before and 24 h after CRT implantation, are summarized in Table 3.

3.2. Implantation

All patients underwent successful implantation. The mean procedure time was 90±30 min. There were no procedure-related complications, deaths or aggravation of haemodynamic status. All left ventricular leads were positioned in a lateral vein either in a proximal or distal site of the vein (lateral in 14, postero-lateral in four and antero-lateral in two) with left ventricular threshold under 3 V and no phrenic stimulation at 5 V. Three patients with a previous implantable cardiac defibrillator (ICD) device for secondary prevention were upgraded to a CRT-defibrillator (CRT-D) device. There were no post-operative lead dislodgments though one patient experienced phrenic stimulation and another had a device pocket haematoma. This haematoma resolved spontaneously and decreased amplitude of left ventricular stimulation successfully avoided diaphragmatic stimulation.

3.3. Haemodynamic status and post-operative period

After CRT implantation, QRS duration slightly decreased compared to spontaneous QRS complexes (Table 3). In all patients, haemodynamic status rapidly improved with successful weaning from intravenous catecholamine mostly within 2 days after implantation (Table 4). Three patients (Patients 9, 15, 16) experienced an immediate increase in both systolic and diastolic blood pressure when the bi-ventricular pacing mode was programmed and could be weaned from dobutamine or epinephrine within 24 h. A marked improvement of both

Table 4
Patient outcomes after CRT-CRT defibrillator implantation

Patient number	Days for catecholamine withdrawal	1-year clinical outcomes	Follow-up	NYHA class after implantation	Treatment after CRT	
					ACE inhibitors	Beta-blockers
Patient 1	2	CV death	7 months	III	Yes	Yes
Patient 2	2	CV death (SCD)	4 months	III	Yes	No
Patient 3	2	CV death (VT)	2 days	N/A	Yes	No
Patient 4	2	Alive	17 months	III	Yes	Yes
Patient 5	5	CV death (SCD)	9 months	III	Yes	Yes
Patient 6	5	CV death	3 months	II	Yes	Yes
Patient 7	2	CV death	8 months	III	Yes	Yes
Patient 8	2	Alive (NSVT)	33 months	II	Yes	Yes
Patient 9	1	Alive (2 DHF, NSVT)	30 months	II	Yes	Yes
Patient 10	2	Alive (1 DHF)	16 months	III	Yes	Yes
Patient 11	1	3 DHF, CV death	9 months	III	Yes	Yes
Patient 12	1	Alive	15 months	II	Yes	Yes
Patient 13	2	Alive	17 months	II	Yes	Yes
Patient 14	2	non-CV death	10 months	II	Yes	Yes
Patient 15	1	Alive (NSVT)	21 months	II	Yes	Yes
Patient 16	1	Alive (NSVT)	21 months	II	Yes	Yes
Patient 17	2	Alive	51 months	II	Yes	Yes
Patient 18	2	CV death	15 days	IV	Yes	No
Patient 19	2	Alive	16 months	II	Yes	Yes
Patient 20	2	Alive	51 months	II	Yes	Yes

CV=cardio-vascular, VT=ventricular tachycardia, NSVT=non-sustained ventricular tachycardia DHF=decompensate heart failure, SCD=sudden cardiac death.

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systolic and diastolic blood pressure (with a mean systolic blood pressure increase of 15 mmHg) was observed in all patients, this was associated with a sharp fall in BNP levels (almost three times lower) (Table 3). This led to a rapid improvement in urine output (over 2 l/day) and in biological markers of renal function. All patients were discharged from the ICU and subsequent hospitalisation in the standard cardiac unit was 8±3 days. One patient died from sudden cardiac death (SCD) 2 days after implantation after spending 36 h in the ICU post-implantation. The cause of death was fatal ventricular tachycardia (VT) during sleep the night following ICU discharge, documented after interrogation of the device memory.

3.4. Follow-up period

The follow-up period was 18±12 months (median: 15 months; minimum: 2 days; maximum: 51 months). No patient experienced late procedure-related complications. All sinus rhythm and AF patients displayed bi-ventricular pacing for more than 90% of the time. Survival rates were 85% at 3 months, 80% at 6 months and 55% at 18 months with no difference between ischaemic and non-ischaemic patients. There were 9 deaths during the follow-up period (including the one occurring at 2 days) which were due to overt HF for 5 patients, SCD for three and severe sepsis for one (Table 4). During the follow-up, all but one patient improved by at least one NYHA class. Eleven patients returned to NYHA class II and remained stable throughout the follow-up period. Three patients had episodes of HF. Among these, one underwent an in-hospital electromechanical dissociation. The three patients who died from SCD (including the one who died from VT at 2 days) had been implanted with a CRT without associated defibrillator. Finally, one patient died from a non-cardiac cause related to a severe sepsis. These last 4 patients died despite a persistent improvement in cardiac haemodynamics. Hence, the rate of hospitalisation for HF during the follow-up was 15%. Echocardiographic assessment of LVEF was performed between 1 and 3 months after CRT implantation and showed slight improvement in LVEF, left ventricular end diastolic and end systolic dimensions (Table 3 and Fig. 1).

4. Discussion

Our retrospective study showed that CRT allowed a rapid and successful catecholamine weaning in all studied CDOHF

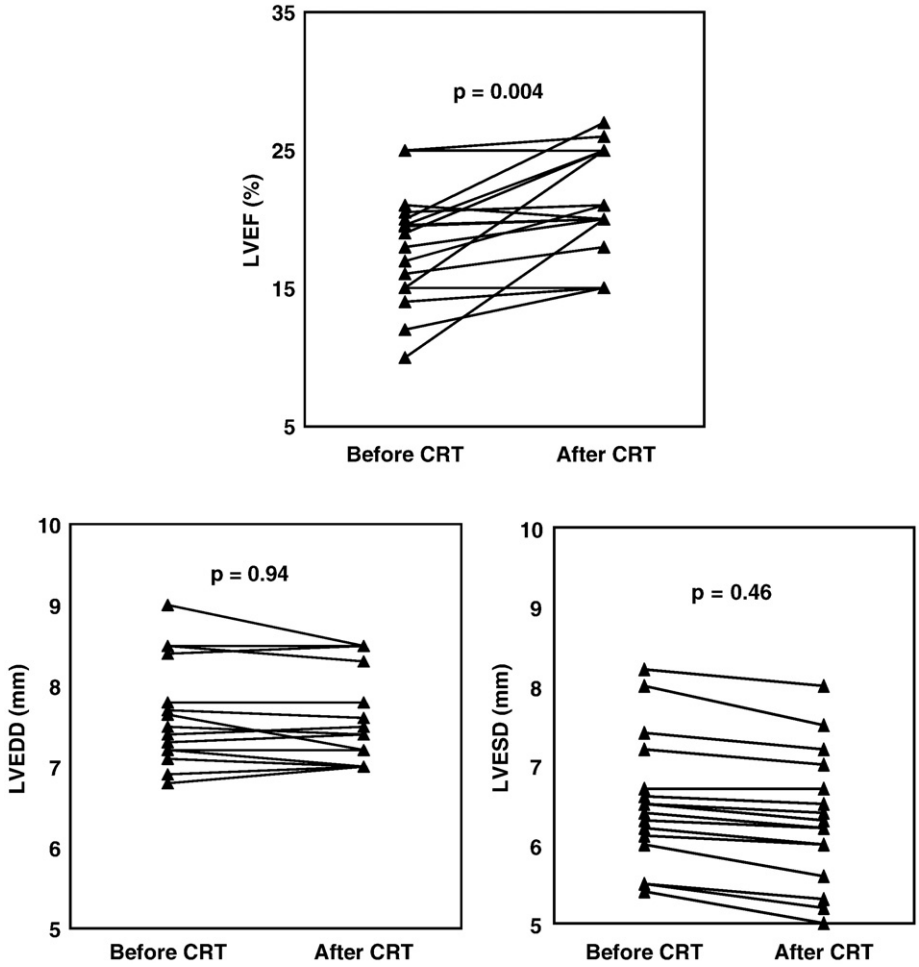

Fig. 1. Individual Echocardiography Data for 17 Patients before and after CRT (3 died before the controlled echocardiography).

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patients. Furthermore, this short-term beneficial effect was sustained for more than one year in surviving patients.

Unstable overt HF is a challenging issue in our clinical practice. When optimal oral medication fails to control HF with persistence of NYHA class IV, intravenous dobutamine or epinephrine and recently levosimendan [17] are effective in managing a critical and severe period of decompensated HF. However, repeated infusions of catecholamine agents for episodes of acute HF appear to be a key point in the management of HF. In such a clinical situation where intravenous drugs fail to control the haemodynamic status of the patient, immediate heart transplantation or implantation of a left VAD is recommended [16]. These options, however, are only possible for eligible patients, i.e. those without severe associated diseases. Because of the limited number of available hearts for transplant, age is also a restrictive criterion for transplantation. To date, no other medical or surgical alternatives are available for such patients. Patients with CDOHF have poor short-term prognosis: in-hospital and one-year mortality stands at 50% or more [18,19]. Our aim here therefore, was to perform what could be considered as "rescue CRT" in these patients who do not fall within the guidelines for CRT indications.

The potential beneficial effects of atrioventricular and then ventriculo-ventricular asynchrony have already been reported in acute haemodynamic studies [9–14]. As a consequence, one might postulate that CRT may be helpful in patients with persistent low cardiac output despite drugs that usually improve cardiac function. It is known that repeat infusions of catecholamine agents in patients with severe overt HF may lead to tolerance due to desensitization of the ß-adrenergic pathway [20–22]. These patients are considered as "non-responders" to these drugs and no haemodynamic improvement is observed. In this setting, because the acute beneficial effects of atrioventricular and ventriculo-ventricular pacing have already been demonstrated; CRT may allow restoration of stable haemodynamics with rapid stopping of intravenous drugs.

Previous reports have already tested this "CRT option" with a short-term follow-up in a small number of inotrope-supported patients [23–25]. Our larger study of consecutive patients confirms that CRT allowed complete catecholamine weaning in CDOHF patients mostly within 2 days following implantation. Improvement of blood pressure after CRT implantation was sustained allowing a marked improvement in renal function. Interestingly, patients who showed an immediate improvement in blood pressure when CRT was turned on, were still alive after 48 months. This suggests that immediate improvement might be a good predictor of long term survival after CRT implantation in CDOHF patients. Finally, a marked drop in BNP levels was observed within 36 h following CRT implantation and this was associated with catecholamine weaning, increased blood pressure and urine output. Improved clinical and biological parameters, with a reduction in BNP levels are indicative of an acute improvement in cardiac function.

Furthermore, persistent improvement in cardiac haemodynamics was observed during the follow-up period with an improved NYHA class, higher LVEF and a low rate of clinical events (death and hospitalisation). During the follow-up, three patients with a CRT device without a defibrillation system died from SCD, this was related to ventricular tachycardia in one patient (two days after device implantation). No ventricular arrhythmia could be documented in the other two patients because the device memory could not be checked. However, it is difficult to know whether the patients would have survived with a CRT-D device, because SCD is not always due to malignant ventricular arrhythmia. In addition, from a cost-effectiveness point of view, it is debatable to systematically implant a CRT-D device in such a clinical setting because of the very poor prognosis of these patients. Because the CARE-HF study showed a reduction of both mortality and SCD in patients with CRT without anti-tachycardia therapies, we considered that a purely haemodynamic device would be a reasonable choice for our patient population. While all patients experienced at least two episodes of decompensated HF within the 15 months before CRT implantation, only three patients had subsequent recurrent episodes of HF. Hence, once the acute unstable haemodynamic phase was controlled by CRT, the clinical improvement of our patients seemed close to what is observed in "stable NYHA class III/IV" CRT patients. We considered that the persistent improvement of our patients was not only related to CRT but also to the medical treatment, such as ACE inhibitors, beta-blockers and spironolactone, which we were able to reintroduce after restoring stable haemodynamics. However, none of these drugs, which are known to improve both morbidity and mortality in NYHA class III/IV patients, could have been given and optimised without prior CRT implantation. Hence, it could be said that mid-term benefits in these patients were due to a combination of both CRT and optimal medical treatment.

There are limitations to our study. Firstly, it is a retrospective and observational report without a systematic biological and echocardiographic protocol. However, we aimed to assess immediate clinical improvement (i.e. weaning from catecholamines; discharge from hospital) and survival after a mean of 1 year follow-up in this severe patient population but not to evaluate reverse remodelling. Of note, the latter is usually observed after 6 months of resynchronisation [3].

Secondly, the number of patients was limited and there was no control group to compare outcomes and thus help select those patients that would benefit most from this expensive and invasive treatment. However, although epidemiological data on the prognosis of such patients exist in the literature [18,19], the use of a "control group", while methodologically interesting, would have been ethically problematic in these catecholamine-dependent class IV patients for whom cardiac transplantation or VAD is not feasible. The common practice is to continue catecholamine treatment until death or to discontinue catecholamine and maintain other supportive therapy either in hospital, at home, or in a hospice

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setting. As a consequence, it is not conceivable to extrapolate our results to a larger study population.

In summary, immediate as well as mid-term clinical improvement was observed after rescue CRT implantation in our CDOHF patients, allowing hospital discharge with persistent functional improvement for most. As a result, our study suggests that CRT might be considered as an additional therapeutic option in CDOHF patients ineligible for left VAD.

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